

(12) **United States Plant Patent**
Freyre et al.

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(54) **RUELLIA PLANT NAMED ‘R15-24-17’**

(50) Latin Name: *Ruellia simplex*
Varietal Denomination: **R15-24-17**

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(57) **ABSTRACT**

‘R15-24-17’ is a new *Ruellia simplex* plant distinguished by having excellent performance under partial sun and in containers, medium-sized bicolor flowers with a white corolla and red-purple corolla tube, average plant height at 47 cm, profuse branching, and no fruit production, as disclosed.

2 Drawing Sheets

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**ACKNOWLEDGMENT OF FEDERAL
RESEARCH SUPPORT**

This invention was made with government support under FLA-ENH-005388 awarded by the National Institute of Food and Agriculture, USDA. The government has certain rights in the invention.

Genus and species: *Ruellia simplex*.
Cultivar denomination: ‘R15-24-17’.

**CROSS-REFERENCE TO RELATED
APPLICATION**

n/a

BACKGROUND OF THE NEW CULTIVAR

The invention relates to a new and distinct variety of *Ruellia simplex* plant named ‘R15-24-17’. ‘R15-24-17’ is an F₁ seedling from a manual hybridization conducted on May 4, 2015, at the Environmental Horticulture Research Greenhouses (EHRG), University of Florida (UF) in Gainesville, Fla., between female parent ‘R14-2x-WPK3’ (UF proprietary clone, not patented; diploid, white corolla with red-purple throat) x male parent ‘R14-4x-WPK3’ (UF proprietary clone, not patented; tetraploid, white corolla with red-purple throat). Ploidy level determinations were done by flow cytometry at UF using wild *R. simplex* as a diploid (2n=2x=34) control and ‘Purple Showers’ (commercial cultivar, not patented) as a tetraploid control. ‘R15-24-17’ is monoploid (n=x=17), which indicates that its origin was likely by haploid parthogenesis in the maternal parent.

‘R15-24-17’ has been reproduced asexually for over two years through vegetative cuttings (stem tip cuttings) and has been shown to retain its distinctive characteristics through successive asexual propagations.

‘R15-24-17’ has not been made publicly available more than one year prior to the filing date of this application.

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SUMMARY OF THE INVENTION

The following are the most outstanding and distinguishing characteristics of ‘R15-24-17’ when grown under normal horticultural practices in Florida.

1. Excellent performance under partial sun environments and in containers;
2. Medium-sized bicolor flowers with a white corolla and red-purple corolla tube;
3. Average plant height of 48 cm;
4. Profuse branching; and
5. No fruit production.

DESCRIPTION OF THE FIGURES

This new *Ruellia simplex* variety ‘R15-24-17’ is illustrated by the accompanying photographs which show the plant’s form and foliage. The colors shown are as true as can be reasonably obtained by conventional photographic procedures. The photographs were taken from seven-week old plants grown at the EHRG, UF, Gainesville, Fla.

FIG. 1 shows a single plant of *Ruellia simplex* ‘R15-24-17’ in a one-gallon container, taken on Sep. 14, 2017, at the EHRG, UF, Gainesville, Fla.; and

FIG. 2 shows a close-up view of a flower of *Ruellia simplex* ‘R15-24-17’ taken on Sep. 14, 2017, at the EHRG, UF, Gainesville, Fla.

**DETAILED BOTANICAL DESCRIPTION OF
THE CULTIVAR**

The following detailed description sets forth the distinctive characteristics of ‘R15-24-17’. Plants were propagated by vegetative cuttings on June 2016, and grown in the EHRG, UF, Gainesville, Fla. Plants were approximately seven weeks of age when plant parts were measured and descriptions of color were determined. Color references are

to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.), 2001 Edition.

BOTANICAL DESCRIPTION

Botanical classification:

Family.—Acanthaceae.

Botanical name.—*Ruellia simplex*.

Common name.—*Ruellia* (*Ruellia*) or Mexican petunia.

Variety name.—‘R15-24-17’.

Plant description:

Form.—Erect.

Habit.—Clumping.

Height (from top of soil).—44-50 cm.

Width (horizontal plant diameter).—46-50 cm.

Propagation:

Type.—Vegetative meristems having at least two nodes.

Time to initiate roots.—7 days.

Time to produce a rooted cutting.—14 days.

Root habit.—Adventitious (fibrous).

Root description.—Callus forms in 4-5 days, roots initiate in 7 days and become a highly branched cutting in 14-21 days.

Branches:

Quantity per plant.—Approximately 40 when not pinched.

Branch color.—RHS 148A (yellow-green) with RHS 187B tint (greyed-purple) when mature. New growth is lighter in color (RHS 138B) and gets darker with light exposure.

Texture.—Smooth.

Pubescence.—Slightly tomentose, especially at nodes, RHS 155D (white).

Stem description.—Round to square shaped stem, 2.0-3.0 cm in diameter at the soil line.

Branch diameter.—1.0 cm at the base of a 45-cm-long branch.

Branch length.—45-50 cm.

Internode length.—2-5 cm.

Anthocyanin.—RHS 183B (greyed-purple).

Leaves:

Quantity of leaves per branch.—Approximately 50 on a 50-cm branch.

Arrangement.—Decussate (opposite pairs).

Fragrance.—None.

Shape.—Lanceolate.

Length.—8-16 cm.

Width.—0.4-1.0 cm.

Apex.—Narrowly acute.

Base.—Attenuate.

Margin.—Entire.

Leaf texture (both surfaces).—Slightly coriaceous.

Venation color.—Upper surface: RHS 138B (green). Lower surface: RHS 138C (green).

Venation pattern.—Pinnate.

Color, mature and immature leaf.—Upper surface: RHS 147A (green) immature, RHS 147A (green) mature. Lower surface: RHS 147B (green) immature, RHS 147A (green) mature.

Petiole length.—1-2 cm.

Petiole diameter.—0.2 cm.

Petiole color.—RHS 146B (yellow-green).

Flowers and seeds:

Flower.—Arrangement: Axillary in solitary or several-flowered cyme. Aspect: Actinomorphic corolla with five elongated petals, funnel form. Flowering habit (length of flowering season): April to October (north Florida). Number of inflorescences per plant: 4-8 cymes with 20-40 flowers plus buds. Fragrance: Very mild and slightly sweet. Lastingness of individual bloom: 1 day. Rate of opening: Daily.

Flower bud.—Shape: Elliptic. Length: 2.0 cm. Diameter: 0.4 cm. Color: RHS 150B (yellow-green), lower surface RHS 71B (red-purple). Texture: Smooth.

Corolla.—Arrangement: Actinomorphic salverform (funnel). Length: 3.0-3.5 cm. Diameter: 3.0-4.0 cm. Color: Upper surface: RHS 155D (white), slightly transparent. Lower surface: RHS 155D (white), slightly transparent. Texture: Rugose.

Corolla tube.—Length: 1.0 cm. Diameter: 0.5 cm. Color: RHS 71B (red-purple), includes central bottom half of lower petal.

Banner.—Not present.

Keel.—Not present.

Calyx.—Number of sepals per flower: 5. Length: 1.0 cm. Diameter: 0.3 cm. Apex: Narrowly acute. Base: Fused. Texture (both surfaces): Smooth to glandular. Pubescence: Present (glandular trichomes). Sepal color: Upper surface: RHS 143B (green). Lower surface: RHS 143B (green).

Pedicels.—Angle: 15-25°. Length: 1.0 cm. Diameter: 0.1 cm. Texture: Smooth to glandular (glandular trichomes). Color: Upper surface: RHS 79A (greyed-purple) under full sun, RHS 143B (green) in partial sun. Lower surface: RHS 143B (green).

Reproductive organs:

Stamens.—Number: 4 fused to the base of the corolla. Filaments: 4. Color: RHS 155D (white). Length: Two at 0.4 cm, two at 0.5 cm. Diameter: 0.1 cm. Anther color: RHS 155D (white). Pollen amount: None observed.

Pistils.—Number per flower: 1. Pistil length: 2.2 cm. Stigma: Color: RHS 155D (white). Shape: Bilabial and slightly recurved. Length: 0.2 cm. Diameter: 0.2 cm. Style: Color: RHS 155D (white). Length: 2.0 cm. Diameter: 0.1 cm. Ovary: Shape: Hypogenous, elliptic. Color: RHS 143A (green). Length: 0.5 cm. Width: 0.1 cm.

Fruit/seed set: None observed.

Disease and insect resistance: Disease and insect resistance is typical of the species, thus no claims are made of any superior disease or insect resistance with this cultivar. The most common insect pests observed in this cultivar, particularly under greenhouse conditions in Gainesville, Fla., have been mealybugs (*Pseudococcidae*), spider mites (*Tetranychidae*), and aphids (*Aphidoidea*). No susceptibility to any particular pathogens has been observed on this cultivar.

COMPARISON WITH KNOWN CULTIVARS

When ‘R15-24-17’ is compared to the female parent ‘R14-2x-WPK3’, ‘R15-24-17’ has shorter plant height and smaller white flowers with a red-purple corolla tube, while ‘R14-2x-WPK3’ has tall plant height and medium white flowers with a red-purple corolla tube. When ‘R15-24-17’ is

compared to the male parent 'R14-4x-WPK3', 'R15-24-17' has shorter plant height and smaller white flowers with a red-purple corolla tube, while 'R14-4x-WPK3' has tall plant height and large white flowers with a red-purple corolla tube. 'R14-2x-WPK3' and 'R14-2x-WPK3' form fruits and seeds while 'R15-24-17' does not. When 'R15-24-17' is compared to the commercial *Ruellia* cultivar Mayan™ White ('R10-108', U.S. Plant Pat. No. 25,156), flowers of 'R15-24-17' are colored white (RHS 155D), with a corolla tube colored red-purple (RHS 71B), whereas flowers of

'R10-108' are colored white (RHS 155C), with a corolla tube also colored white (RHS 155C). Additionally, when compared to 'R10-108', plants of 'R15-24-17' are more compact having a height of 44-50 cm and a width of 46-50 cm, whereas plants of 'R10-108' have a height of 62.7 cm and a width of 60.6 cm.

We claim:

1. A new and distinct variety of *Ruellia* plant named 'R15-24-17' as illustrated and described herein.

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FIG. 1



FIG. 2